Section: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3180**

Title: **INTER-RIVET BUCKLING**

Revision: **1.0**

Date: **26/06/2019**

1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

Calculation of the inter-rivet buckling load in a fastened joint between a stringer and the skin in a metallic stiffened panel loaded in longitudinal compression.

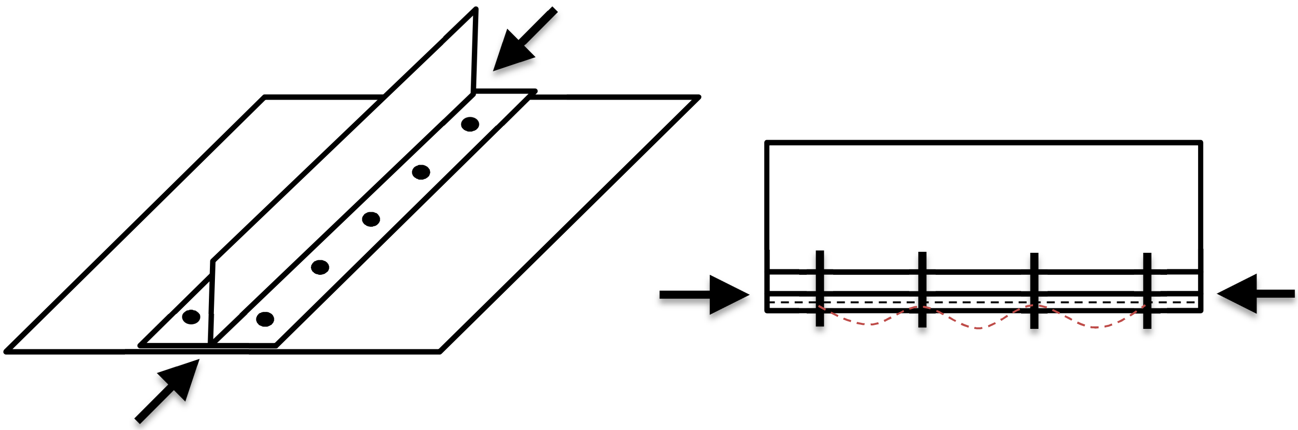

Figure 1. Longitudinal fastened joint with by-pass compression and potential inter-rivet buckling

2 METHOD DESCRIPTION

In a stiffened panel loaded in longitudinal compression, if the stringer is riveted to the skin, the panel may suffer inter-rivet buckling. Even the stringer foot flange may have this failure also, the flange and the fastened joint should be sized to retard inter-rivet buckling beyond the flange crippling strength.

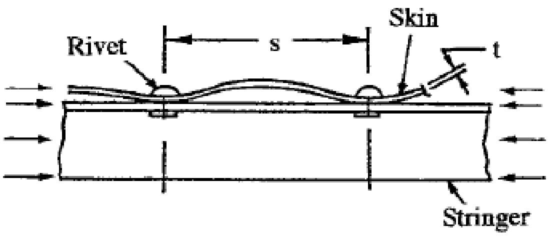

Figure 2. Inter-rivet buckling

The expression to calculate of the inter-rivet buckling load can be found in literature, e.g. Bruhn's chapter C7.14 (formula C7.21):

$$F_{ir} = \frac{\pi^2 E_t}{\left(\frac{\frac{p}{\sqrt{C}}}{0.29t}\right)^2}$$

…where:

- E_t is the tangent modulus of the material
- p is the pitch, or distance, between rivets
- C is the fixity (or clamping) coefficient at rivets positions. Typical values for C are:

  C = 1.5: countersunk rivets

  C = 3.0: universal rivets

  C = 4.0: flat head rivets
- t is the thickness

Beyond the material linear limit, as the tangent modulus is function of the stress in the material, this equation is transcendental and shall be solved numerically.

If the inter-rivet buckling stress calculates to be more than the crippling stress of the stiffener, then the effective sheet area can be added to the stiffener area to obtain the total effective area. This total effective area times the stiffener crippling stress will give the crippling load for the total sheet stiffener unit.

When the sheet between rivets buckles before the crippling stress of the stiffener is reached, the sheet in the buckled state has the ability to approximately hold this stress as the stiffener continues to take load until it reaches the stiffener crippling stress. This buckling sheet strength can be taken advantage off by reducing the effective sheet area. Thus effective width equals,

$w' = w \cdot (F_{ir} / F_{st})$ (Bruhn, formula C7.23)

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
C	-	Fixity coefficient
E_t	MPa	Tangent modulus
F_{st}	MPa	Crippling stress of the stiffener
F_{ir}	MPa	Inter-rivet buckling stress
p	mm	Rivets pitch
t	mm	Thickness

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[M. C. Y. Niu]	2^{nd} Ed.	AIRFRAME STRESS ANALYSIS AND SIZING

SOFTWARE

Program	***Version***	***Description***
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures

Table 1. *Software programs referenced in this chapter*

DISTRIBUTION LIST

Department	***Name***	***Full Doc.***	***Cover Page***
Stress Methods & Optimisation (TEASP-TL3)	Fernass Daoud		X
Fatigue & Damage Tolerance GET (TEASA-TL1)	Javier Gomez-Escalonilla Martin		
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
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Fatigue & Damage Tolerance MAN (TEASA-TL7)	Patrick Schneider		

Table 2. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.1 – 1.0 21/10/2018	G. Baños	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Analysis (TEASA)	Miguel Ángel Calero 26/06/2019	Bernhard Mayer 26/06/2019
Stress Methods (TEASP-TL3)	G. Baños 21/10/2018	Florian Glock 26/06/2019

Table 4. *Approval signatures table for last revision of this chapter*